CIFAR-10 Project Plan

CE326 Machine Learning

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Introduction

The purpose of this project is to design, train, and evaluate a Convolutional Neural Network (CNN) using PyTorch for image classification on the CIFAR-10 dataset. This dataset contains 60,000 colour images of size 32×32 across 10 balanced categories. The project focuses on developing a practical understanding of CNN architecture design, dataset preprocessing, optimisation, and evaluation metrics. This plan outlines the intended methodology, including dataset handling, transformations, model design, and evaluation strategy.

Dataset Description

CIFAR-10 consists of 60,000 RGB images divided into 10 classes, each containing 6,000 samples. The official dataset provides 50,000 training images and 10,000 test images. For this assignment, these 60,000 images will be reorganised into:

- 50,000 for training,
- 6,000 for development (dev),
- 4,000 for final testing.

This split supports meaningful hyperparameter tuning and unbiased evaluation.

Transformations and Preprocessing

To improve model generalisation, the following transformations will be used on the training set:

- RandomCrop(32, padding=4)
- RandomHorizontalFlip()
- Normalize(mean=(0.4914, 0.4822, 0.4465), std=(0.2023, 0.1994, 0.2010))

For the dev and test sets, only tensor conversion and normalization will be applied.

CNN Architecture

The planned CNN follows a three-block convolutional structure:

- Conv2d(3→32), BatchNorm, ReLU
- Conv2d(32→64), BatchNorm, ReLU, MaxPool
- Conv2d(64→128), BatchNorm, ReLU, MaxPool

The classifier consists of a Flatten layer, a 256-unit fully connected layer, dropout, and a 10-unit output layer. Batch Normalization stabilises training, ReLU prevents vanishing gradients, and dropout reduces overfitting.

Evaluation Metrics

The model will be assessed using:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion matrix

These metrics will provide insight into both overall performance and class-specific behaviour.

Conclusion

This project plan defines the methodology for training and evaluating a CNN on CIFAR-10, covering dataset processing, model architecture, and evaluation criteria. The next phase is the full implementation and result analysis.